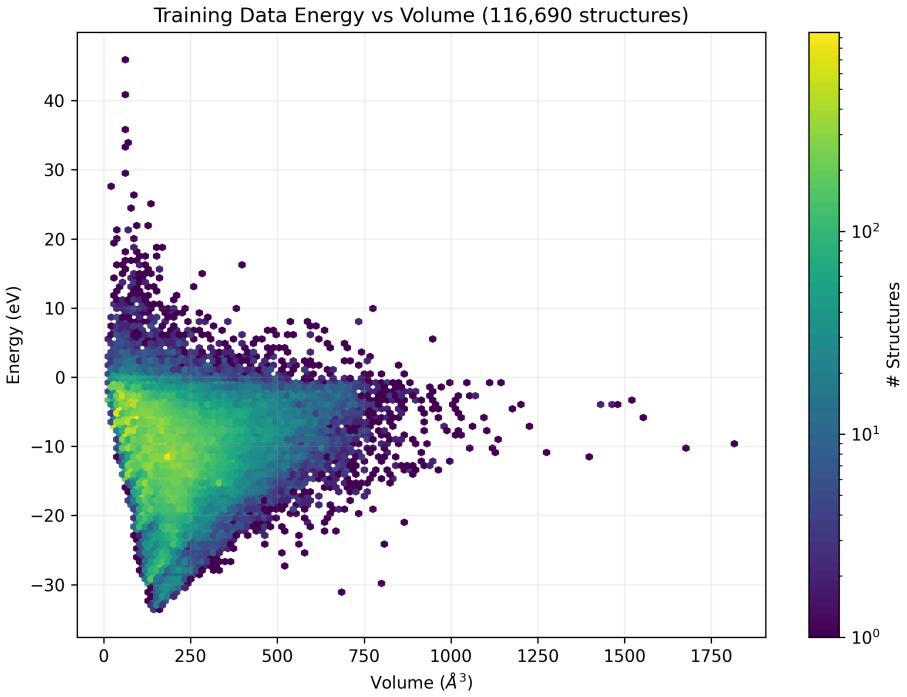
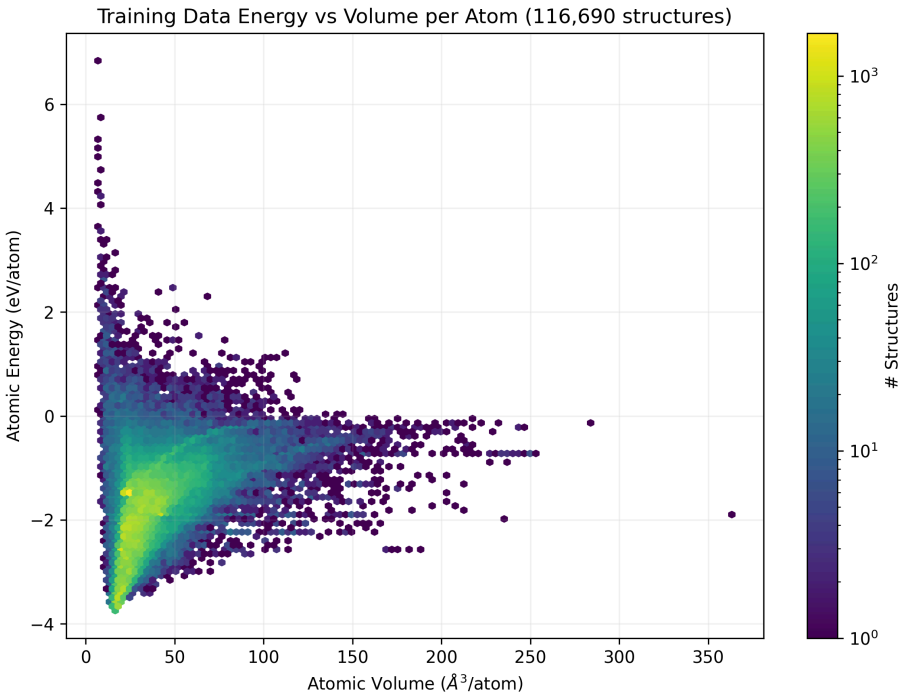
Requested Information

Plots





Structures that have volume of 200 or more per atom

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11117 Al4
cell: Cell([[6.44882112, 0.01504105, -0.02768478], [0.01504105, 6.45954048,
0.29885216], [-0.00701874, 0.07576509, 22.86751788]])
positions: [[ 7.52052500e-03  3.22977024e+00  1.49426080e-01]
 [ 3.22441056e+00  7.52052500e-03 -1.38423900e-02]
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 [ 3.22090119e+00  4.54030700e-02  1.14199165e+01]
 [ 4.01115500e-03  3.26765279e+00  1.15831850e+01]]
forces: [[-0.  0.  0.]
 [-0. -0.  0.]
 [ 0.  0. -0.]
 [-0. -0. -0.]]
metadata: {'name': 11116}
11226 Al6
cell: Cell([[5.55597273, -3.20836502, -0.48095676], [-0.00053867, 5.6601383,
-0.28622166], [-0.05627763, -0.00618005, 38.85833683]])
positions: [[ 3.68973304e+00 -2.53742328e-01  9.29853874e+00]
 [ 1.80942339e+00  2.69933556e+00  2.87926197e+01]
 [-4.61902345e-02  5.65512500e+00  3.12351517e+01]
 [-3.87644916e-02 -4.25683347e-03  2.67661318e+01]
 [ 5.54534709e+00 -3.20953177e+00  6.85575176e+00]
 [ 5.53792055e+00  2.44985002e+00  1.13252815e+01]]
forces: [[ 0.20590271  0.11194645  0.02016097]
 [-0.20592814 -0.11194775 -0.02016379]
 [-0.04611291 -0.01579288 -0.61053914]
 [ 0.25204903  0.12771752  0.63076661]
 [ 0.04611312  0.01578394  0.61049146]
 [-0.2520238  -0.12770728 -0.63071611]]
metadata: {'name': 11225}
11832 Ca
cell: Cell([[6.32239558, -0.06637916, 0.10579459], [-0.06637944, 6.2285992,
0.00101643], [0.10583529, 0.00101392, 5.85940949]])
positions: [[ 6.32239539 -0.06637903  0.10579459]]
forces: [[ 0.  0. -0.]]
metadata: {'name': 11831}
12000 Ca2
cell: Cell([[10.20822713, 0.02738017, -0.06463589], [0.04505802, 6.52934332,
0.08662482], [-0.11424576, 0.0913754, 6.28128587]])
positions: [[10.15110372  0.0730715   3.07600967]
 [ 5.06952031  3.32405106  3.15162699]]
forces: [[-0. -0.  0.]
 [ 0.  0. -0.]]
metadata: {'name': 11999}
15298 Ca3
cell: Cell([[10.12706578, -5.6313129, -0.04743179], [0.18674632, 10.96065652,
-0.14401331], [-0.0774493, -0.25810462, 5.81333834]])
positions: [[ 5.02467723 -2.94465844  2.88780645]
 [ 0.05493017  5.35109667  2.83449018]
 [ 5.11804375  2.53575832  2.8062687 ]]
forces: [[-5.0990e-05 -5.5000e-07 -2.8942e-04]
 [ 1.0728e-04 -1.5910e-05  3.9800e-06]
 [-5.6300e-05  1.6450e-05  2.8544e-04]]
metadata: {'name': 15297}
15967 Ca4
cell: Cell([[10.66242294, -5.99491583, 0.05404223], [0.13967372, 12.68440613,
0.15002578], [0.09310709, 0.23595333, 5.97616078]])
positions: [[ 5.51261734  9.80759193  3.1657108 ]
 [10.77870026  0.45953507  3.11683282]
 [ 5.45263679  3.46580888  3.08983928]
 [10.80200867  6.68947581  0.20406903]]
forces: [[-2.0521e-04  6.7370e-05  9.8000e-07]
 [-1.0544e-04 -1.8439e-04  5.4200e-06]
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 [ 2.2568e-04  1.8506e-04 -7.0000e-06]
 [ 8.4970e-05 -6.8040e-05  5.9000e-07]]
metadata: {'name': 15966}
16081 Ca3
cell: Cell([[10.21325754, -5.99986956, 0.0759962], [-0.05960812, 12.19086926,
0.07051534], [0.13283786, 0.0655398, 5.97899234]])
positions: [[ 5.04704683  9.19083294  0.10851421]
 [10.18336145  0.09557008  0.11126025]
 [ 5.2097284   3.1611473   6.05224288]]
forces: [[ 4.600e-06 -2.279e-05 -5.000e-07]
 [-1.620e-05  5.030e-06  4.600e-07]
 [ 1.160e-05  1.776e-05  5.000e-08]]
metadata: {'name': 16080}
16183 Ca4
cell: Cell([[14.16988138, -0.00739167, 0.09461379], [-0.00738646, 9.34003953,
-0.04173439], [0.2025261, -0.08968603, 6.36513492]])
positions: [[ 3.36844569  4.59169164  5.44222934]
 [11.14033622  4.58760844  5.49394248]
 [ 7.15512353  2.072811    2.29704283]
 [ 7.1511271   7.19617396  2.27399633]]
forces: [[ 2.5524741e-01  3.0446000e-04 -2.2344700e-02]
 [-2.5524741e-01 -3.0446000e-04  2.2344700e-02]
 [ 1.9528000e-04  2.0785750e-01  5.2629600e-03]
 [-1.9528000e-04 -2.0785750e-01 -5.2629600e-03]]
metadata: {'name': 16182}
16522 Ca4
cell: Cell([[5.83501271, -0.32671596, 1.02141193], [-0.06507179, 6.79568043,
0.07758347], [9.58982833, 12.2049929, 22.61023456]])
positions: [[ 1.36102105  8.60098273  3.43519573]
 [11.98782578 14.3823584  15.76117279]
 [ 3.82372001  4.86601902  9.01304791]
 [14.44975259 10.64752419 21.33754934]]
forces: [[ 3.8190e-05 -5.9119e-04 -3.9310e-04]
 [-2.3330e-05 -3.5147e-04 -3.4961e-04]
 [ 1.1660e-05  3.4659e-04  4.0978e-04]
 [-2.6520e-05  5.9606e-04  3.3293e-04]]
metadata: {'name': 16521}
17549 Ca
cell: Cell([[5.5947361, 0.04253502, -0.10223161], [0.04253502, 6.16227407,
0.0538416], [-0.10223161, 0.0538416, 6.04607213]])
positions: [[0. 0. 0.]]
forces: [[0. 0. 0.]]
metadata: {'name': 17548}
17558 Ca3
cell: Cell([[7.8305014, 0.22963502, 0.14050088], [0.22963466, 8.1667242,
-0.06800105], [0.1002348, -0.04852873, 11.44249282]])
positions: [[ 0.050118   -0.02425634  5.72124406]
 [ 8.06013512  8.39635364  0.07249986]
 [ 4.0801902   4.17391754  5.75748918]]
forces: [[ 0.  0.  0.]
 [ 0. -0.  0.]
 [-0.  0.  0.]]
metadata: {'name': 17557}
19084 Ca3
cell: Cell([[8.11241776, -4.43297002, -0.03307842], [0.21619475, 11.96447016,
0.03227476], [-0.05202508, 0.08859414, 6.17261958]])
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positions: [[4.12557619 3.83171675 4.59693372]
 [4.25097929 9.78443123 2.55548818]
 [0.10403354 5.98918047 0.49835162]]
forces: [[-3.896700e-04  2.955320e-03  5.079431e-02]
 [ 3.519000e-04 -2.080880e-03 -9.012240e-03]
 [ 3.777000e-05 -8.744400e-04 -4.178206e-02]]
metadata: {'name': 19083}
19464 Ca3
cell: Cell([[10.16424425, -5.96214428, -0.02824327], [-0.0812479, 11.41021423,
0.09054174], [-0.04923299, 0.21048228, 6.28564917]])
positions: [[ 5.08212283 -2.98107199 -0.01412163]
 [-0.04062323  5.70510001  0.04527547]
 [ 4.99227715  2.93451896  6.31679379]]
forces: [[-0. -0.  0.]
 [-0.  0. -0.]
 [ 0. -0.  0.]]
metadata: {'name': 19463}
19579 Ca4
cell: Cell([[8.00035752, -4.72272891, 0.1461857], [-0.08986034, 10.88321948,
0.27499736], [0.10655007, 0.16999149, 13.13594769]])
positions: [[ 2.6601642   5.76623114  6.80117289]
 [ 5.30362792  0.47929017  0.19024859]
 [ 5.35688234  0.56424916  6.7559574 ]
 [ 2.713418    5.85120666 13.36687134]]
forces: [[-0.07186006 -0.03833545 -0.00233883]
 [ 0.07188461  0.03837618  0.00260412]
 [ 0.07186006  0.03833545  0.00233883]
 [-0.07188461 -0.03837618 -0.00260412]]
metadata: {'name': 19578}
20746 Ca4
cell: Cell([[5.89702988, 0.11870908, 0.06171213], [0.11870905, 5.90598639,
-0.04098906], [0.01539611, -0.01028719, 25.47029916]])
positions: [[ 5.49255152e-05  5.46389519e-05  8.81201295e-04]
 [ 6.02337929e+00  6.01948136e+00  1.27551237e+01]
 [ 5.90086861e+00  1.16146862e-01  6.42804228e+00]
 [ 1.15503841e-02 -7.70859172e-03  1.91038494e+01]]
forces: [[ 3.3200e-06 -4.6200e-06  8.3260e-05]
 [-4.8200e-06  5.0300e-06 -5.4280e-05]
 [-3.6300e-06  6.1900e-06 -1.2327e-04]
 [ 5.1300e-06 -6.6000e-06  9.4290e-05]]
metadata: {'name': 20745}
34724 Al3Ca
cell: Cell([[5.20280634, -2.84588474, -0.16964106], [0.13683528, 8.59596622,
0.04589455], [-0.07499219, 0.07055745, 17.99784449]])
positions: [[ 2.58324371 -1.40591001  4.19177512]
 [ 0.05123575  4.31451018  4.29969226]
 [ 2.6512474   2.89133152  4.21378246]
 [-0.03014027  0.02922276  7.34181813]]
forces: [[ 0.02792635 -0.01296352  0.50524488]
 [ 0.00362747 -0.00362403  0.07258206]
 [ 0.00618833  0.00622712  0.19192705]
 [-0.03774215  0.01036042 -0.76975399]]
metadata: {'name': 34723}
42272 Mg2
cell: Cell([[8.6726447, 0.0, 2.07921573], [0.0, 10.92179491, 0.0], [-1.02148678, 0.0,
4.24927983]])
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positions: [[6.24914286 8.74937509 2.62180908]
 [1.40201506 8.74937509 3.70668648]]
forces: [[ 1.851e-05  0.000e+00  6.190e-06]
 [-1.851e-05  0.000e+00 -6.190e-06]]
metadata: {'name': 42271}
43302 Mg
cell: Cell([9.01066522, 4.76108423, 5.04702298])
positions: [[0.         0.         2.52351149]]
forces: [[0. 0. 0.]]
metadata: {'name': 43301}
43878 Mg
cell: Cell([[5.31941287, -2e-08, 0.0], [-2.43732578, 4.2215721, 0.0], [0.0, 0.0,
9.18417357]])
positions: [[0.         0.         1.03799144]]
forces: [[0. 0. 0.]]
metadata: {'name': 43877}
54193 CaMg
cell: Cell([[4.05040336, 0.08091019, 0.16466091], [0.0809227, 6.69836588,
0.12593888], [0.07478167, 0.05725232, 16.7441132]])
positions: [[2.10305963 3.41825995 8.51735364]
 [4.08778615 0.10953693 8.53671837]]
forces: [[-5.08e-06  3.67e-06  1.04e-06]
 [ 5.08e-06 -3.67e-06 -1.04e-06]]
metadata: {'name': 54192}
54483 CaMg2
cell: Cell([[6.98541835, -0.0197419, 0.23397819], [-0.01898881, 5.10802454,
0.03381714], [0.08159976, 0.01225354, 19.34615995]])
positions: [[3.48410667 5.09971339 2.61416387]
 [6.98577072 2.53574976 2.58624635]
 [3.51076439 2.54827779 6.66363996]]
forces: [[-0.00035816 -0.00151723  0.21521422]
 [-0.00938501 -0.00040586  0.10481379]
 [ 0.00974317  0.00192309 -0.32002801]]
metadata: {'name': 54482}
58473 CaMg2
cell: Cell([[7.79031045, -4.42470285, 0.11004238], [0.06325076, 8.96519955,
-0.07596161], [0.08156955, -0.11211, 8.83529743]])
positions: [[ 5.25540775 -0.01745728  4.46568958]
 [ 7.85356011  4.5404957   0.03408103]
 [ 5.2961955  -0.07351091  8.8833384 ]]
forces: [[-8.08000e-06 -6.01000e-05 -1.40400e-05]
 [-3.36011e-03  6.62887e-03  8.03000e-06]
 [ 3.36819e-03 -6.56877e-03  6.01000e-06]]
metadata: {'name': 58472}
73541 AlMg2
cell: Cell([[7.76521106, -15.49428284, -1e-07], [7.76521087, 15.49428275, -9e-08],
[-4e-08, 0.0, 2.93324381]])
positions: [[ 7.76521086e+00  1.19111192e+01  2.55185476e+00]
 [ 7.76521100e+00 -1.19111193e+01  2.55185479e+00]
 [ 7.76521094e+00 -4.50000002e-08  1.64240284e+00]]
forces: [[-0.00000e+00  6.71189e-03 -2.08000e-05]
 [ 0.00000e+00 -6.71189e-03 -2.08000e-05]
 [-0.00000e+00  0.00000e+00  4.16100e-05]]
metadata: {'name': 73540}
73542 AlMg2
cell: Cell([[7.76517033, -15.49419499, -1.1e-07], [7.76517014, 15.49419489, -9e-08],
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[-4e-08, 0.0, 2.93288731]])
positions: [[ 7.76517013e+00  1.19110523e+01  2.55154462e+00]
 [ 7.76517027e+00 -1.19110524e+01  2.55154463e+00]
 [ 7.76517021e+00 -5.00000006e-08  1.64220319e+00]]
forces: [[-0.00000e+00  6.72095e-03 -2.36700e-05]
 [ 0.00000e+00 -6.72095e-03 -2.36700e-05]
 [-0.00000e+00  0.00000e+00  4.73500e-05]]
metadata: {'name': 73541}
73543 AlMg2
cell: Cell([[7.76523395, -15.49433224, -1e-07], [7.76523377, 15.49433215, -9e-08],
[-4e-08, 0.0, 2.93344426]])
positions: [[ 7.76523376e+00  1.19111569e+01  2.55202915e+00]
 [ 7.76523389e+00 -1.19111570e+01  2.55202914e+00]
 [ 7.76523384e+00 -4.50000002e-08  1.64251508e+00]]
forces: [[-0.0000e+00  6.7018e-03 -7.4800e-06]
 [ 0.0000e+00 -6.7018e-03 -7.4800e-06]
 [-0.0000e+00  0.0000e+00  1.4960e-05]]
metadata: {'name': 73542}
73544 AlMg2
cell: Cell([[7.76497634, -15.49377649, -1.1e-07], [7.76497615, 15.49377639, -9e-08],
[-4e-08, 0.0, 2.93118913]])
positions: [[ 7.76497614e+00  1.19107334e+01  2.55006730e+00]
 [ 7.76497636e+00 -1.19107336e+01  2.55006731e+00]
 [ 7.76497622e+00 -4.99999997e-08  1.64125222e+00]]
forces: [[-0.00000e+00  6.80875e-03 -4.08100e-05]
 [ 0.00000e+00 -6.80875e-03 -4.08100e-05]
 [-0.00000e+00  0.00000e+00  8.16200e-05]]
metadata: {'name': 73543}
73545 AlMg2
cell: Cell([[7.76388492, -15.49142194, -1.3e-07], [7.7638847, 15.49142183, -1.1e-07],
[-4e-08, 0.0, 2.92163489]])
positions: [[ 7.76388461e+00  1.19089393e+01  2.54175557e+00]
 [ 7.76388494e+00 -1.19089394e+01  2.54175558e+00]
 [ 7.76388479e+00 -5.50000001e-08  1.63590196e+00]]
forces: [[-0.00000e+00  7.27789e-03 -2.27500e-05]
 [ 0.00000e+00 -7.27789e-03 -2.27500e-05]
 [-0.00000e+00  0.00000e+00  4.55000e-05]]
metadata: {'name': 73544}
73546 AlMg2
cell: Cell([[7.77166846, -15.50821351, 0.0], [7.77166846, 15.50821351, 0.0], [0.0,
0.0, 2.98977149]])
positions: [[  7.77166846  11.92173412   2.601031  ]
 [  7.77166846 -11.92173412   2.601031  ]
 [  7.77166846   0.           1.67405761]]
forces: [[ 0.00000e+00  4.26421e-03  4.98200e-05]
 [ 0.00000e+00 -4.26421e-03  4.98200e-05]
 [-0.00000e+00  0.00000e+00 -9.96500e-05]]
metadata: {'name': 73545}
73718 AlMg2
cell: Cell([[9.8474864, -12.97084117, -4e-08], [9.84756211, 12.97089863, 5e-08],
[0.0, 1e-08, 2.93314029]])
positions: [[ 4.92378106e+00  6.48544932e+00  2.50000000e-08]
 [ 4.92374320e+00 -6.48542058e+00  9.33140290e-09]
 [ 0.00000000e+00  4.99999990e-09  1.46657012e+00]]
forces: [[-0. -0.  0.]
 [ 0.  0.  0.]
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 [-0. -0.  0.]]
metadata: {'name': 73717}
73719 AlMg2
cell: Cell([[9.84879781, -12.97558007, 0.0], [9.84879781, 12.97558007, 0.0], [0.0,
0.0, 2.97879055]])
positions: [[ 4.92439891  6.48779003  0.        ]
 [ 4.92439891 -6.48779003  0.        ]
 [ 0.          0.          1.48939527]]
forces: [[-0.  0.  0.]
 [ 0.  0.  0.]
 [-0. -0.  0.]]
metadata: {'name': 73718}
74849 AlMg2
cell: Cell([[15.17943757, -1e-08, -1e-08], [-1e-08, 15.17943134, 1e-08], [0.0, 0.0,
2.93713054]])
positions: [[-4.99999800e-09  7.58971263e+00  4.99999800e-09]
 [ 7.58971575e+00 -4.99999800e-09 -4.99999800e-09]
 [ 0.00000000e+00  0.00000000e+00  1.46856216e+00]]
forces: [[ 0.  0.  0.]
 [-0.  0.  0.]
 [ 0.  0.  0.]]
metadata: {'name': 74848}
75588 AlMg3
cell: Cell([[18.20669664, -2.41e-05, 1e-08], [-9.10336919, 15.76745674, -1e-08],
[0.0, 0.0, 2.96146934]])
positions: [[ 9.10334832e+00 -1.20500000e-05  5.00000000e-09]
 [-4.55168460e+00  7.88372837e+00 -5.00000000e-09]
 [ 4.55166373e+00  7.88371632e+00  0.00000000e+00]
 [ 0.00000000e+00  0.00000000e+00  0.00000000e+00]]
forces: [[-0.  0. -0.]
 [-0.  0.  0.]
 [ 0.  0. -0.]
 [-0.  0. -0.]]
metadata: {'name': 75587}
75589 AlMg3
cell: Cell([[18.21193102, -1.568e-05, 1e-08], [-9.10597909, 15.77199162, -1e-08],
[0.0, 0.0, 2.99084164]])
positions: [[ 9.10596551e+00 -7.84000000e-06  5.00000000e-09]
 [-4.55298954e+00  7.88599581e+00 -5.00000000e-09]
 [ 4.55297597e+00  7.88598797e+00  0.00000000e+00]
 [ 0.00000000e+00  0.00000000e+00  0.00000000e+00]]
forces: [[ 0.  0.  0.]
 [-0.  0.  0.]
 [ 0.  0.  0.]
 [ 0.  0.  0.]]
metadata: {'name': 75588}
75590 AlMg3
cell: Cell([[18.2216734, 0.0, 0.0], [-9.1108367, 15.78043206, 0.0], [0.0, 0.0,
3.0455102]])
positions: [[ 9.1108367   0.          0.        ]
 [-4.55541835  7.89021603  0.        ]
 [ 4.55541835  7.89021603  0.        ]
 [ 0.          0.          0.        ]]
forces: [[-0.  0.  0.]
 [ 0.  0.  0.]
 [-0.  0.  0.]
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 [-0.  0.  0.]]
metadata: {'name': 75589}
75598 AlMg4
cell: Cell([[18.65815789, -4.6e-07, -4.98e-06], [-4.6e-07, 18.65816188, 1.55e-06],
[-8.1e-07, 2.5e-07, 3.01134763]])
positions: [[ 6.01642722e+00  1.86581612e+01 -5.58289843e-08]
 [ 1.26417300e+01  6.14928228e-08 -3.37417098e-06]
 [ 6.14927154e-08  1.26417338e+01  1.05019378e-06]
 [ 1.86581574e+01  6.01642776e+00 -4.45008029e-06]
 [ 9.32907890e+00  9.32908071e+00 -1.71500005e-06]]
forces: [[ 0.01059785 -0.         -0.        ]
 [-0.01059785  0.          0.        ]
 [ 0.         -0.01059785 -0.        ]
 [-0.          0.01059785  0.        ]
 [-0.         -0.         -0.        ]]
metadata: {'name': 75597}
75599 AlMg4
cell: Cell([18.67872557, 18.67872557, 3.04575768])
positions: [[ 6.02286241  0.          0.        ]
 [12.65586316  0.          0.        ]
 [ 0.         12.65586316  0.        ]
 [ 0.          6.02286241  0.        ]
 [ 9.33936279  9.33936279  0.        ]]
forces: [[ 0.00875031 -0.          0.        ]
 [-0.00875031  0.         -0.        ]
 [-0.         -0.00875031  0.        ]
 [ 0.          0.00875031 -0.        ]
 [-0.          0.         -0.        ]]
metadata: {'name': 75598}
75649 AlMg2
cell: Cell([[9.84748751, -12.97084516, -4e-08], [9.84756315, 12.97090258, 5e-08],
[0.0, 1e-08, 2.93317876]])
positions: [[ 4.92378157e+00  6.48545129e+00  2.50000000e-08]
 [ 4.92374376e+00 -6.48542258e+00  9.33178760e-09]
 [ 0.00000000e+00  4.99999990e-09  1.46658935e+00]]
forces: [[-0. -0.  0.]
 [ 0.  0.  0.]
 [-0.  0.  0.]]
metadata: {'name': 75648}
75650 AlMg2
cell: Cell([[9.84746068, -12.97074823, -4e-08], [9.84753787, 12.97080682, 5e-08],
[0.0, 1e-08, 2.93224499]])
positions: [[ 4.92376894e+00  6.48540341e+00  2.50000000e-08]
 [ 4.92373034e+00 -6.48537411e+00  9.32244990e-09]
 [ 0.00000000e+00  4.99999990e-09  1.46612247e+00]]
forces: [[-0. -0.  0.]
 [ 0.  0.  0.]
 [-0. -0.  0.]]
metadata: {'name': 75649}
75720 AlMg2
cell: Cell([[9.84748205, -12.97082543, -4e-08], [9.847558, 12.97088308, 5e-08], [0.0,
1e-08, 2.93298864]])
positions: [[ 4.92377900e+00  6.48544154e+00  2.50000000e-08]
 [ 4.92374102e+00 -6.48541271e+00  9.32988640e-09]
 [ 0.00000000e+00  4.99999990e-09  1.46649429e+00]]
forces: [[-0. -0.  0.]
```

```
 [ 0.  0.  0.]
 [ 0. -0.  0.]]
metadata: {'name': 75719}
75721 AlMg2
cell: Cell([[9.84748964, -12.97085287, -4e-08], [9.84756516, 12.97091019, 5e-08],
[0.0, 1e-08, 2.93325296]])
positions: [[ 4.92378258e+00  6.48545509e+00  2.50000000e-08]
 [ 4.92374482e+00 -6.48542643e+00  9.33252960e-09]
 [ 0.00000000e+00  4.99999990e-09  1.46662645e+00]]
forces: [[-0. -0.  0.]
 [ 0.  0.  0.]
 [-0. -0.  0.]]
metadata: {'name': 75720}
75722 AlMg2
cell: Cell([[9.84727366, -12.97007242, -4e-08], [9.84736165, 12.9701392, 6e-08],
[0.0, 1e-08, 2.92573484]])
positions: [[ 4.92368082e+00  6.48506960e+00  3.00000000e-08]
 [ 4.92363683e+00 -6.48503621e+00  9.25734840e-09]
 [ 0.00000000e+00  4.99999990e-09  1.46286739e+00]]
forces: [[ 0.  0.  0.]
 [ 0. -0.  0.]
 [ 0.  0.  0.]]
metadata: {'name': 75721}
76089 AlMg2
cell: Cell([[14.91545167, -1.4e-07, -2.6e-07], [-1.4e-07, 14.91540688, 1.5e-07],
[-5e-08, 3e-08, 2.93056159]])
positions: [[7.45773299e+00 9.92525802e-06 1.89068216e-01]
 [1.49154316e+01 7.45772284e+00 1.89068044e-01]
 [9.85131371e-06 1.49153869e+01 1.19513902e-01]]
forces: [[ 0.00e+00 -0.00e+00 -2.18e-06]
 [ 0.00e+00 -0.00e+00 -2.19e-06]
 [-0.00e+00  0.00e+00  4.37e-06]]
metadata: {'name': 76088}
76220 AlMg3
cell: Cell([[18.2077159, -2.246e-05, 1e-08], [-9.1038774, 15.76833979, -1e-08], [0.0,
0.0, 2.96718884]])
positions: [[ 9.10385795e+00 -1.12300000e-05  5.00000000e-09]
 [-4.55193870e+00  7.88416990e+00 -5.00000000e-09]
 [ 4.55191925e+00  7.88415867e+00  0.00000000e+00]
 [ 0.00000000e+00  0.00000000e+00  0.00000000e+00]]
forces: [[-0.  0.  0.]
 [ 0.  0.  0.]
 [ 0.  0.  0.]
 [-0.  0.  0.]]
metadata: {'name': 76219}
76259 AlMg3
cell: Cell([[12.03933223, 5e-08, -0.00038801], [1e-08, 2.97224481, 5e-08],
[-1.0160467, 3.6e-07, 22.65284816]])
positions: [[ 6.01966610e+00  1.55020281e+00 -1.93525865e-04]
 [ 1.02961584e-01  1.22790449e+00  6.30040613e+00]
 [ 1.09203241e+01  1.22790470e+00  1.63520545e+01]
 [ 1.10232855e+01  7.50522482e-01  2.26524593e+01]]
forces: [[ 0.000000e+00  2.549000e-04  0.000000e+00]
 [-6.238000e-04  1.170000e-06 -4.020805e-02]
 [ 6.238000e-04  1.170000e-06  4.020805e-02]
 [-0.000000e+00 -2.572400e-04 -0.000000e+00]]
```

```
metadata: {'name': 76258}
76339 Al2Mg4
cell: Cell([[14.35481111, 1.77e-06, 1.66e-06], [4.21e-06, 34.13287123, 4.11e-06],
[3.5e-07, 3.6e-07, 2.91705793]])
positions: [[3.05028389e-06 2.10649557e+01 2.57161663e+00]
 [7.17740716e+00 1.30679189e+01 1.11308710e+00]
 [7.17740837e+00 2.10649555e+01 1.80397637e+00]
 [1.65326455e-06 1.30679183e+01 3.45445670e-01]
 [4.55999979e-06 3.41328699e+01 2.91706201e+00]
 [7.17740994e+00 3.41328709e+01 1.45853390e+00]]
forces: [[-0.00000e+00 -1.39253e-03  3.92900e-05]
 [ 0.00000e+00  1.39253e-03  3.92900e-05]
 [-0.00000e+00 -1.39253e-03 -3.92900e-05]
 [ 0.00000e+00  1.39253e-03 -3.92900e-05]
 [ 0.00000e+00  0.00000e+00  0.00000e+00]
 [ 0.00000e+00  0.00000e+00  0.00000e+00]]
metadata: {'name': 76338}
76340 Al2Mg4
cell: Cell([[14.36157435, 1.72e-06, 1.67e-06], [4.1e-06, 34.14845251, 4.09e-06],
[3.5e-07, 3.6e-07, 2.92114465]])
positions: [[2.98246697e-06 2.10745890e+01 2.57521424e+00]
 [7.18078873e+00 1.30738668e+01 1.11464141e+00]
 [7.18078992e+00 2.10745888e+01 1.80650873e+00]
 [1.61114905e-06 1.30738659e+01 3.45934791e-01]
 [4.44999979e-06 3.41484512e+01 2.92114871e+00]
 [7.18079145e+00 3.41484522e+01 1.46057725e+00]]
forces: [[-0.         -0.00128937  0.00010935]
 [ 0.          0.00128937  0.00010935]
 [-0.         -0.00128937 -0.00010935]
 [ 0.          0.00128937 -0.00010935]
 [ 0.         -0.          0.        ]
 [ 0.          0.         -0.        ]]
metadata: {'name': 76339}
76359 Al2Mg4
cell: Cell([[25.58933361, -2.6e-07, -0.00209331], [-3e-08, 2.92991492, -2e-08],
[-4.20051073, -1e-07, 19.54337079]])
positions: [[12.99736356  2.25242814 10.82208173]
 [15.0976157   0.67748641  1.05039479]
 [-3.37048108  1.15753597 18.88171396]
 [-1.27022503  1.77237879  9.11002895]
 [ 7.38211677  1.47314459  3.91306084]
 [ 5.28186253  1.45677004 13.68474936]]
forces: [[-9.73322e-03 -5.83000e-06 -9.76540e-04]
 [-9.73322e-03  5.83000e-06 -9.76540e-04]
 [ 9.21263e-03 -2.52200e-05  3.99428e-03]
 [ 9.21263e-03  2.52200e-05  3.99428e-03]
 [ 5.20590e-04 -3.24000e-05 -3.01774e-03]
 [ 5.20590e-04  3.24000e-05 -3.01774e-03]]
metadata: {'name': 76358}
76360 Al2Mg4
cell: Cell([[25.58799091, -2.9e-07, -0.00232023], [-3e-08, 2.92307869, -2e-08],
[-4.20046367, -1.1e-07, 19.54289833]])
positions: [[12.99657769  2.2471724  10.821684  ]
 [15.09680603  0.67590588  1.05023328]
 [-3.3704641   1.15483447 18.8812553 ]
 [-1.27023158  1.76824405  9.10980652]
```

```
 [ 7.38169478  1.46970728  3.91288939]
 [ 5.28146428  1.4533711  13.68434188]]
forces: [[-9.77137e-03  1.03500e-05 -9.82970e-04]
 [-9.77137e-03 -1.03500e-05 -9.82970e-04]
 [ 9.25012e-03  4.38400e-05  3.99658e-03]
 [ 9.25012e-03 -4.38400e-05  3.99658e-03]
 [ 5.21240e-04 -4.95000e-06 -3.01362e-03]
 [ 5.21240e-04  4.95000e-06 -3.01362e-03]]
metadata: {'name': 76359}
76361 Al2Mg4
cell: Cell([[3.01163581, 0.0, 0.0], [0.0, 22.98894738, 0.0], [-0.73886856, 0.0,
21.48466962]])
positions: [[-0.36943428  0.         10.74233481]
 [ 0.          0.          0.        ]
 [ 1.13638363  8.09251891 10.74233481]
 [ 1.13638363 14.89642847 10.74233481]
 [ 1.20729555 11.49447369 17.28664639]
 [ 1.0654717  11.49447369  4.19802323]]
forces: [[-0.          0.          0.        ]
 [-0.          0.          0.        ]
 [-0.          0.0119749   0.        ]
 [-0.         -0.0119749  -0.        ]
 [-0.00013383  0.         -0.02024847]
 [ 0.00013383 -0.          0.02024847]]
metadata: {'name': 76360}
85828 Al2Mg2
cell: Cell([[3.15101896, 0.0, 0.0], [-1.57550948, 2.72886247, 0.0], [0.0, 0.0,
96.0718956]])
positions: [[ 0.         0.         0.       ]
 [ 0.         0.        48.0359478]
 [ 0.         0.        76.2601299]
 [ 0.         0.        19.8117657]]
forces: [[ 0.          0.          0.        ]
 [ 0.          0.          0.        ]
 [-0.          0.         -0.00020697]
 [-0.         -0.          0.00020697]]
metadata: {'name': 85827}
85861 Al2Mg2
cell: Cell([[3.25574266, 0.0, 0.0], [-1.62787133, 2.81955585, 0.0], [0.0, 0.0,
89.0599561]])
positions: [[ 0.          0.         66.46851157]
 [ 0.          0.         22.59144453]
 [ 0.          0.         76.02564322]
 [ 0.          0.         13.03431288]]
forces: [[-0.          0.          0.00072998]
 [ 0.         -0.         -0.00072998]
 [ 0.          0.         -0.00046635]
 [-0.         -0.          0.00046635]]
metadata: {'name': 85860}
88447 AlMg2
cell: Cell([15.1934462, 15.1934462, 2.96342628])
positions: [[0.         7.5967231  0.        ]
 [7.5967231  0.         0.        ]
 [0.         0.         1.48171314]]
forces: [[0. 0. 0.]
 [0. 0. 0.]
```

```
 [0. 0. 0.]]
metadata: {'name': 88446}
88448 AlMg2
cell: Cell([14.74681633, 14.74681633, 2.87631276])
positions: [[0.         7.37340816 0.        ]
 [7.37340816 0.         0.        ]
 [0.         0.         1.43815638]]
forces: [[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
metadata: {'name': 88447}
88456 AlMg2
cell: Cell([14.96714409, 14.96714409, 2.95971071])
positions: [[7.48357204 0.         0.1909233 ]
 [0.         7.48357204 0.1909233 ]
 [0.         0.         0.12075691]]
forces: [[-0.00e+00 -0.00e+00 -1.27e-06]
 [-0.00e+00  0.00e+00 -1.27e-06]
 [ 0.00e+00  0.00e+00  2.55e-06]]
metadata: {'name': 88455}
88457 AlMg2
cell: Cell([14.54104853, 14.54104853, 2.87545151])
positions: [[7.27052426 0.         0.18548796]
 [0.         7.27052426 0.18548796]
 [0.         0.         0.11731911]]
forces: [[-0.00e+00  0.00e+00  1.58e-06]
 [ 0.00e+00 -0.00e+00  1.58e-06]
 [-0.00e+00  0.00e+00 -3.16e-06]]
metadata: {'name': 88456}
88487 AlMg3
cell: Cell([[3.14573599, 0.0, 0.0], [-1.572868, 2.72428728, 0.0], [0.0, 0.0,
99.44919186]])
positions: [[ 0.          0.          4.02976181]
 [ 0.          0.         41.80021979]
 [ 0.          0.         74.1984543 ]
 [ 0.          0.         64.83295992]]
forces: [[-0.0000e+00  0.0000e+00 -3.8500e-06]
 [ 0.0000e+00 -0.0000e+00  8.3250e-05]
 [-0.0000e+00 -0.0000e+00 -6.6915e-04]
 [ 0.0000e+00 -0.0000e+00  5.8975e-04]]
metadata: {'name': 88486}
89167 AlMg2
cell: Cell([[7.48353618, -14.93325114, 0.0], [7.48353618, 14.93325114, 0.0], [0.0,
0.0, 2.87892661]])
positions: [[  7.48353618  11.4797394    2.50459855]
 [  7.48353618 -11.4797394    2.50459855]
 [  7.48353618   0.           1.61199242]]
forces: [[ 0.000000e+00  1.209486e-02  1.575000e-05]
 [ 0.000000e+00 -1.209486e-02  1.575000e-05]
 [ 0.000000e+00  0.000000e+00 -3.150000e-05]]
metadata: {'name': 89166}
90849 AlMg2
cell: Cell([[9.54253912, -12.57209081, 0.0], [9.54253912, 12.57209081, 0.0], [0.0,
0.0, 2.88616193]])
positions: [[ 4.77126956  6.28604541  0.        ]
 [ 4.77126956 -6.28604541  0.        ]
```

```
 [ 0.          0.          1.44308097]]
forces: [[-0. -0.  0.]
 [ 0.  0.  0.]
 [ 0. -0.  0.]]
metadata: {'name': 90848}
91261 Al4Mg2
cell: Cell([[131.00925828, 0.0, 0.0], [0.0, 1.67884502, 2.90784487], [0.0,
-1.67884502, 2.90784487]])
positions: [[ 27.58473167   0.           1.93856131]
 [103.42452661   0.           1.93856131]
 [ 74.00618019   0.           1.93856131]
 [ 57.00307809   0.           1.93856131]
 [ 65.50462914   0.           0.        ]
 [  0.           0.           0.        ]]
forces: [[-0.00029353 -0.         -0.        ]
 [ 0.00029353 -0.         -0.        ]
 [-0.00143382  0.          0.        ]
 [ 0.00143382  0.          0.        ]
 [ 0.         -0.          0.        ]
 [ 0.          0.          0.        ]]
metadata: {'name': 91260}
91278 Al4Mg
cell: Cell([[3.4822815, 0.0, 0.0], [-1.74114075, 3.01574424, 0.0], [0.0, 0.0,
172.94993228]])
positions: [[  0.           0.          87.41134993]
 [  0.           0.         128.16149129]
 [  0.           0.          41.05457475]
 [  0.           0.          26.12814986]
 [  0.           0.          15.36615181]]
forces: [[ 0.         -0.          0.00033269]
 [-0.          0.          0.00016387]
 [-0.         -0.         -0.00016268]
 [ 0.         -0.         -0.00076055]
 [ 0.         -0.          0.00042667]]
metadata: {'name': 91277}
91289 AlMg3
cell: Cell([[12.27295035, 0.0, 0.0], [0.0, 3.05319616, 0.0], [-1.03501888, 0.0,
23.07816774]])
positions: [[ 6.13647518  1.59240722  0.        ]
 [ 0.10517833  1.26134687  6.4194116 ]
 [11.13275314  1.26134687 16.65875614]
 [ 0.          0.77098052  0.        ]]
forces: [[ 0.000000e+00  1.559700e-04  0.000000e+00]
 [-5.605500e-04  2.212000e-05 -3.136801e-02]
 [ 5.605500e-04  2.212000e-05  3.136801e-02]
 [ 0.000000e+00 -2.002100e-04  0.000000e+00]]
metadata: {'name': 91288}
91290 AlMg3
cell: Cell([[12.07447241, 0.0, 0.0], [0.0, 3.00381993, 0.0], [-1.01828057, 0.0,
22.70494802]])
positions: [[ 6.03723621  1.56665484  0.        ]
 [ 0.10347739  1.24094839  6.31559699]
 [10.95271445  1.24094839 16.38935103]
 [ 0.          0.75851223  0.        ]]
forces: [[ 0.000000e+00  2.708700e-04  0.000000e+00]
 [-7.428500e-04  9.490000e-06 -3.712691e-02]
```

```
 [ 7.428500e-04  9.490000e-06  3.712691e-02]
 [ 0.000000e+00 -2.898500e-04  0.000000e+00]]
metadata: {'name': 91289}
91315 Al4Mg2
cell: Cell([[3.22066223, 0.0, 0.0], [0.0, 156.48256553, 0.0], [-1.61033111, 0.0,
2.78917531]])
positions: [[ 1.61033268e+00  1.00320746e+02  9.29724146e-01]
 [ 1.61033268e+00  1.34403102e+02  9.29724146e-01]
 [-1.55868785e-06  5.61618197e+01  1.85945116e+00]
 [-1.55868785e-06  2.20794631e+01  1.85945116e+00]
 [ 1.61033268e+00  3.91206414e+01  9.29724146e-01]
 [-1.55868785e-06  1.17361924e+02  1.85945116e+00]]
forces: [[ 0.          0.00012425 -0.        ]
 [ 0.         -0.00012425 -0.        ]
 [-0.         -0.00012425  0.        ]
 [-0.          0.00012425  0.        ]
 [-0.          0.         -0.        ]
 [-0.          0.         -0.        ]]
metadata: {'name': 91314}
```

Code

```python
#!/usr/bin/env python3
from __future__ import annotations

import argparse
import os
from pathlib import Path

os.environ.setdefault(
    "MPLCONFIGDIR", str((Path(__file__).resolve().parent / ".matplotlib").resolve())
)

import matplotlib

matplotlib.use("Agg")

import matplotlib.pyplot as plt
from ase.db import connect
from plot_EvV_FvV_RAW_Data import plot_energy_volume_density


def load_structures(db_path: Path, limit: int | None = None):
    db = connect(str(db_path))
    structures = []

    for index, row in enumerate(db.select(), start=1):
        if limit is not None and index > limit:
            break
        structures.append(row.toatoms())

    return structures


def parse_args():
    parser = argparse.ArgumentParser(
        description="Plot energy-vs-volume views for the training data."
```

```python
    )
    parser.add_argument(
        "--db",
        type=Path,
        default=Path("TrainingData/ase.db"),
        help="Path to the ASE database.",
    )
    parser.add_argument(
        "--output-energy-volume",
        type=Path,
        default=Path("TrainingData/energy_vs_volume.png"),
        help="Where to save the total energy vs total volume figure.",
    )
    parser.add_argument(
        "--output-energy-volume-per-atom",
        type=Path,
        default=Path("TrainingData/energy_vs_volume_per_atom.png"),
        help="Where to save the energy per atom vs volume per atom figure.",
    )
    parser.add_argument(
        "--limit",
        type=int,
        default=None,
        help="Optional number of structures to plot for a quicker preview.",
    )
    return parser.parse_args()


def main():
    args = parse_args()
    structures = load_structures(args.db, args.limit)

    if not structures:
        raise SystemExit("No structures were loaded from the training database.")

    args.output_energy_volume.parent.mkdir(parents=True, exist_ok=True)
    plt.figure(figsize=(8, 6))
    plot_energy_volume_density(structures)
    plt.xlabel(r"Volume ($\AA^3$)")
    plt.ylabel("Energy (eV)")
    plt.title(f"Training Data Energy vs Volume ({len(structures):,} structures)")
    plt.grid(alpha=0.15)
    plt.tight_layout()
    plt.savefig(args.output_energy_volume, dpi=300)
    plt.close()

    args.output_energy_volume_per_atom.parent.mkdir(parents=True, exist_ok=True)
    plt.figure(figsize=(8, 6))
    plot_energy_volume_density(structures, per_atom=True)
    plt.xlabel(r"Atomic Volume ($\AA^3$/atom)")
    plt.ylabel("Atomic Energy (eV/atom)")
    plt.title(
        f"Training Data Energy vs Volume per Atom ({len(structures):,} structures)"
    )
    plt.grid(alpha=0.15)
    plt.tight_layout()
```

```python
    plt.savefig(args.output_energy_volume_per_atom, dpi=300)
    plt.close()

    print(f"Saved energy-vs-volume plot to {args.output_energy_volume}")
    print(
        "Saved energy-vs-volume-per-atom plot to "
        f"{args.output_energy_volume_per_atom}"
    )


if __name__ == "__main__":
    main()
```